Curriculum Vitae

NAFIS ASLAM

nafisaslam1819@gmail.com nafisaslam.com github.com/NafisAslam70

Research Interests: Deep Learning · Computer Vision · Multimodal AI · Vision-Language Systems · Data Science and AI

System Demo: Real-Time Attention Monitoring (YOLO + MediaPipe)

Publication: Aslam, N. (2026). DeepLens Engine for Focus: A Multi-Layered Framework for Real-Time Attention Monitoring in Productivity Environments. *Zenodo Preprint.* DOI: 10.5281/zenodo.19266394

Education

Universiti Sains Malaysia — Penang, Malaysia

BSc (Hons.) Computer Science — Major: Intelligent Computing (AI)

2021 – 2025

- CGPA: 3.32/4.00 Final 4 Semesters: 3.49 → 3.65 → 3.73 → 3.82
- Dean's List — 3 consecutive final semesters
- Gold Award — FYP DeepWork AI (PIXEL 2025, top 5%)

Massachusetts Institute of Technology — MITx / edX (online)

MicroMasters in Statistics and Data Science

Aug 2024 – Dec 2025

- 16-month program (4 graduate-level courses, 10–15 hrs/week) · self-funded
- Data Analysis (93%), Machine Learning (83%), Probability (78%), Statistics (60%), Capstone (73%)

Research & Major Projects

2024–2025

DeepWork AI / DeepLens Engine for Focus (DLEF) *Gold Award, PIXEL 2025*

Stack: YOLOv11n-cls, MediaPipe (gaze, EAR), OpenCV, PyTorch, Next.js

Designed a multi-layer computer vision framework classifying six attention states (Focused, Phone, Absent, Drowsy, Looking Away, Bad Posture) in real time.

Developed a novel **20/30s temporal heuristic** that models attention as a time-series signal rather than per-frame classification — suppressing false positives from momentary distraction and producing interpretable, actionable output.

Deployed as a privacy-preserving web application: on-device inference (<50ms latency), no video storage, with Pomodoro timer, real-time focus overlay, and session analytics.

Authored preprint (DOI: 10.5281/zenodo.19266394); won Gold Award (top 5%) at USM PIXEL 2025.

Research-Adjacent Experience

2026–present	**Deep Learning and Computer Vision Research Fellow** — WorldQuant University (online) Developing 6 applied CV projects (classification, detection, GANs, diffusion models) on cloud VMs. Building end-to-end pipelines for experimentation, model benchmarking, and reproducibility; deploying via Flask/Streamlit.
Mar 2026–present	**Machine Learning Intern** — Unified Mentor (Remote) Built time-series forecasting models (ARIMA, Random Forest, Gradient Boosting) on 3+ years of weekly government care load data. Developed unsupervised learner segmentation (K-Means, Hierarchical clustering) with feature engineering from clickstream logs; co-authored internal research reports.
Oct 2025–Feb 2026	**Software Engineer** — Meed Public School, India (Freelance) Built full-stack platform (Next.js, Drizzle, NeonDB, Twilio) for 30+ users; integrated LLM assistant for administrative Q&A with offline inference.
Mar 2023–Aug 2023	**Software Engineering Intern** — USM Speech Processing Group End-to-end stock analysis application (React, Node.js); real-time data processing, interactive visualisation, and data pipeline construction.

Technical Skills

Computer Vision	YOLO (v9/v11), OpenCV, MediaPipe, real-time detection, gaze estimation, ByteTrack (familiar), optical flow, temporal heuristics
Deep Learning	PyTorch, CNNs, Transformers, Transfer Learning, GANs, Diffusion Models
Machine Learning	Scikit-learn, Pandas, Ensemble Methods, Time-Series (ARIMA, GBM), Clustering, statistical analysis
Algorithms & Optimisation	Model pruning, latency optimisation, heuristic design, cross-validation, hyperparameter tuning
MLOps & Deployment	Docker, Flask, Streamlit, ONNX (basic), Git, cloud VMs (Azure/GCP), on-device inference
Other Tools	Jupyter, Linux, Hugging Face, PostgreSQL, Weights & Biases (basic), CUDA (basic), Next.js

Achievements

- **Gold Honor (Top 2%)** — International Youth Math Challenge 2020
- **Competitive Programming:** Active on LeetCode (∼50 problems solved); active on Kaggle
- **Leadership:** Superintendent, Meed Public School (2018–2022) — directed academics and operations for 300+ residential students; structural attention gaps observed here directly motivated the DLEF research
- **Languages:** English (fluent), Hindi (native), Urdu (native), Bahasa Malaysia (basic)

References available upon request · Last updated: April 2026